

TECHNICAL DOCUMENTATION

Isaac Sim · Digital Twin · AWS EC2 · Amazon DCV

Complete Issue Log, Troubleshooting Guide & Operations Manual
Hospital Simulation — Operating Theatre (OT) & Patient Room

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Component	Technology	Status
Cloud Server	Amazon EC2 (ap-south-1)	■ Running
Remote Desktop	Amazon DCV (port 8443)	■ Active
Server Management	AWS SSM Session Manager	■ Connected
3D Simulation	NVIDIA Isaac Sim / Omniverse	■ Operational
Hospital Environment	Digital Twin — OT + Patient Room	■ Live
Robot Platform	Boston Dynamics Spot	■ Sensor Config

Prepared by the Technical Team for internal reference and CEO review.

● 1. Project Overview

This document records every technical issue, error, and resolution encountered during the setup and operation of the Isaac Omniverse Hospital Simulation project on Amazon Web Services (AWS). It serves as both a troubleshooting reference and a complete operations manual for anyone — technical or non-technical — who works with this system.

1.1 What We Built

We created a fully detailed, photorealistic 3D hospital simulation environment using NVIDIA Isaac Omniverse. The simulation includes:

- Operating Theatre (OT) — with surgical equipment, operating tables, anaesthesia machines, instrument tables, and medical monitors
- Patient Room — with hospital beds, nurse stations, storage cabinets, medical carts, and patient monitoring equipment
- Cloud Infrastructure — the simulation runs on a powerful Amazon EC2 cloud server accessible from anywhere in the world via a secure web browser
- Robot Integration — a Boston Dynamics Spot robot is being configured with sensors inside the simulation

1.2 What is a Digital Twin?

A digital twin is an exact virtual replica of a real-world environment. In our case, the hospital simulation is a digital twin of a real hospital. It allows the team to:

- Test medical robots safely without risking real patients or equipment
- Train staff on procedures in a risk-free virtual environment
- Study workflow and movement patterns before implementing changes in real hospitals
- Predict problems and fix them before they happen in the real world

1.3 Technology Stack

Technology	What It Does	Why We Use It
NVIDIA Isaac Sim	3D simulation platform	Photorealistic physics-based simulation
NVIDIA Omniverse	Collaboration & rendering framework	USD file format, ray tracing, materials
Amazon EC2	Cloud server (Ubuntu, ap-south-1)	Powerful GPU compute in the cloud
Amazon DCV	Remote desktop via browser	Access simulation from anywhere
AWS SSM	Server terminal via AWS console	Manage server without SSH keys
Boston Dynamics Spot	Quadruped robot platform	Real-world robot being simulated

● 2. Complete Issue Log & Resolutions

The following section documents every issue encountered, its root cause in plain language, and the exact steps taken to resolve it.

Issue 1 — Amazon DCV: 'No Session Available' Error

Field	Detail
When	First access attempt via browser
Where	Browser → https://13.207.37.221:8443
Severity	High — complete access blocked

What Happened

When we opened the DCV URL in the browser, the page showed the DCV logo and the error: 'No session is available or you are not authorised to join the session.'

■ *Error: No session is available or you are not authorised to join the session. Please contact the owner of the session or the host administrator.*

Why It Happened

Amazon DCV works like a TV channel — the server can broadcast a desktop, but someone has to start the broadcast first. A DCV session is that broadcast. When the EC2 instance started, no one had created a session yet, so there was nothing to connect to. It is like calling a video call number where no one has started the call yet.

How We Fixed It

We connected to the server via EC2 Instance Connect (a separate terminal method) and ran the command to create a new session:

```
dcv create-session mysession --owner ubuntu
```

```
dcv list-sessions
```

■ Session 'mysession' appeared as Active. DCV browser access was immediately restored.

Prevention

■ Every time the EC2 instance restarts, a new DCV session must be created manually. This is normal behaviour — DCV sessions do not persist across reboots.

Issue 2 — SSM Agent Offline: No IAM Role Attached

Field	Detail
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When	SSM Session Manager connection attempt
Where	AWS Console → EC2 → Connect → SSM Session Manager
Severity	High — SSM access blocked

What Happened

The SSM Session Manager page showed Ping Status: Offline and Session Manager connection status: Not connected. The Connect button was greyed out and unusable.

■ *Error: DHMC is not enabled and IAM instance profile is not attached*

Why It Happened

Think of an IAM role as an ID card that proves your server's identity to AWS. Without this ID card, the server cannot communicate with AWS services including SSM. Our EC2 instance had no IAM role attached at all — it had no ID card — so when the SSM agent tried to connect to AWS, it was turned away at the door with an access denied error.

How We Fixed It

- Went to AWS Console → EC2 → Instances → selected the instance
- Clicked Security → Modify IAM role
- Used the DHMC (Default Host Management Configuration) setup
- Searched for and selected the EC2-SSM-Role
- Clicked Update IAM role

■ ■ After attaching the role, the SSM agent needed 1–2 minutes to detect and use the new role. Refreshing the page too quickly showed it still offline.

Issue 3 — SSM Role Could Not Be Assumed (Repeated Offline)

Field	Detail
When	After attaching EC2-SSM-Role
Where	SSM agent logs on the server
Severity	Critical — kept repeating every restart

What Happened

Even after attaching the EC2-SSM-Role, the SSM agent kept going offline repeatedly. The same error appeared every 5–10 minutes. Restarting the agent had no lasting effect.

■ *Error: ValidationException: Could not assume role arn:aws:iam::867344428658:role/EC2-SSM-Role — status code: 400*

Why It Happened

The EC2-SSM-Role had two internal configuration problems that prevented it from working:

- Problem 1 — Missing Trust Policy: A trust policy is a rule that says which AWS services are allowed to use a role. The EC2-SSM-Role had no rule allowing EC2 servers to use it. It was like giving someone a key card that the door reader was not programmed to accept.
- Problem 2 — Missing Permission Policy: The role also lacked the AmazonSSMManagedInstanceCore permission — the specific permission that allows a server to communicate with AWS Systems Manager. Without this, even a correctly trusted role cannot connect to SSM.

How We Fixed It

Went to IAM Console → Roles → EC2-SSM-Role and made two changes:

Fix A — Updated Trust Policy:

```
{ "Version": "2012-10-17",  
  "Statement": [{  
    "Effect": "Allow",  
    "Principal": { "Service": "ec2.amazonaws.com" },  
    "Action": "sts:AssumeRole"  
  }]  
}
```

Fix B — Attached Permission Policy:

- Clicked Add permissions → Attach policies
- Searched for AmazonSSMManagedInstanceCore
- Selected it and clicked Add permissions

■ After both fixes, the SSM agent connected permanently. This issue has not recurred since.

Issue 4 — SSM Agent Service Not Found via systemctl

Field	Detail
When	Trying to restart SSM agent via terminal
Where	EC2 Instance Connect terminal
Severity	Medium — caused confusion, quickly resolved

What Happened

When trying to restart the SSM agent using the standard Linux command, the terminal returned a 'not found' error:

```
Error: Failed to restart amazon-ssm-agent.service: Unit amazon-ssm-agent.service not found.
```

Why It Happened

On most Linux systems, services are managed by a tool called systemctl. However, our server runs Ubuntu, and on Ubuntu, AWS installs the SSM agent through a completely different package manager called Snap. These two systems are independent — systemctl has no knowledge of anything installed through Snap. It is exactly like trying to find an app in the App Store that was only ever available on Google Play.

How We Fixed It

Used the correct Snap commands instead:

```
sudo snap install amazon-ssm-agent --classic # install if missing
```

```
sudo snap start amazon-ssm-agent # start the agent
```

```
sudo snap restart amazon-ssm-agent # restart the agent
```

```
sudo snap services amazon-ssm-agent # check status
```

```
Output showed: Service enabled, Status active — agent running correctly.
```

```
On Ubuntu, always use 'snap' commands for the SSM agent, never 'systemctl'.
```

Issue 5 — DCV Session Lost After New Instance Launch

Field	Detail
When	After launching a new EC2 instance
Where	Browser → https://13.202.35.102:8443
Severity	Medium — same error as Issue 1, expected behaviour

What Happened

After a new EC2 instance was launched with a new IP address (13.202.35.102), the DCV browser showed the same 'No session available' error again.

Why It Happened

DCV sessions are temporary and live only on the specific server instance they were created on. When a new instance is launched, it is a completely fresh server with no sessions. Additionally, the IP address changed, so the old URL no longer pointed to the right server. This is expected behaviour — not a bug.

How We Fixed It

- Checked the new IP address in EC2 console
- Connected via EC2 Instance Connect to the new instance
- Created a fresh DCV session:

```
dcv create-session mysession --owner ubuntu
```

- Accessed the simulation at the new IP:

```
https://13.202.35.102:8443/?sessionId=mysession
```

■ Hospital simulation was fully accessible again within 2 minutes.

● 3. Issue Summary Table

#	Issue	Root Cause	Fix	Status
1	DCV No Session Error	Session not created	dcv create-session command	■ Fixed
2	SSM Agent Offline	No IAM role attached to EC2	Attached EC2-SSM-Role via AWS console	■ Fixed
3	SSM Role Cannot Be Assumed	Wrong trust policy + missing permission	Fixed IAM trust policy + added policy	■ Fixed
4	SSM systemctl Not Found	Ubuntu uses Snap, not systemctl	Used correct snap commands	■ Fixed
5	DCV Lost After New Instance	Sessions don't persist across reboots	Created new session on new instance	■ Fixed

● 4. Operations Manual — Step-by-Step Procedures

This section is for anyone who needs to operate the system day-to-day. Follow each procedure in order.

4.1 Starting the System (Every Session)

Step	Action	Where	Expected Result
1	Log in to AWS Console	https://console.aws.amazon.com	AWS dashboard opens
2	Go to EC2 → Instances	Left menu → Instances	Instance list appears
3	Start Isaac-sim-server	Select instance → Instance state → Start	Status: Running (green)
4	Copy new Public IP address	Click instance → Public IPv4 address	IP like 13.x.x.x
5	Open EC2 Instance Connect	Connect → EC2 Instance Connect → Connect	Terminal opens in browser
6	Create DCV session	Type command in terminal	Session created successfully
7	Open DCV in browser	https://YOUR-IP:8443/?sessionId=mysessionId	Remote desktop appears
8	Launch Isaac Sim	Terminal in DCV desktop: ./isaac-sim.sh	Isaac Sim loads (3–5 min)
9	Open hospital environment	File → Open → hospital USD file	3D hospital scene loads

4.2 Shutting Down Properly

■ Always shut down properly to protect your work and avoid unnecessary AWS costs.

Step	Action	Why Important
1	Save your work in Isaac Sim (File → Save)	Prevents losing any changes to the scene
2	Close Isaac Sim	Releases GPU resources on the server
3	Go to EC2 → Instances in AWS Console	Access the server controls
4	Select instance → Instance state → Stop instance	Stops AWS billing for compute time
5	Wait for status: Stopped	Confirms server is fully off

■ Never click 'Terminate' — that permanently deletes the server and all its data.

4.3 Quick Reference Commands

Task	Command
Create DCV session	<code>dcv create-session mysession --owner ubuntu</code>
List DCV sessions	<code>dcv list-sessions</code>
Close DCV session	<code>dcv close-session mysession</code>
Restart SSM agent	<code>sudo snap restart amazon-ssm-agent</code>
Check SSM agent status	<code>sudo snap services amazon-ssm-agent</code>
View SSM agent logs	<code>sudo snap logs amazon-ssm-agent -n 50</code>
Launch Isaac Sim	<code>./isaac-sim.sh</code> (in Isaac Sim install folder)
Check disk space	<code>df -h</code>
Check GPU usage	<code>nvidia-smi</code>

4.4 Accessing the Server (Three Methods)

Method	How to Use	Best For	Requires
EC2 Instance Connect	AWS Console → Connect → EC2 Instance Connect → Connect	Quick terminal access, running commands	AWS Console access
Amazon DCV Browser	<code>https://YOUR-IP:8443/?sessionId=mysession</code>	Full remote desktop, running Isaac Sim	DCV session to be created first
SSM Session Manager	AWS Console → Connect → SSM Session Manager → Connect	Secure terminal, no SSH keys needed	IAM role + SSM agent running

● 5. Troubleshooting Guide

Use this table if you encounter any issues. Find your error and follow the fix.

Problem You See	Most Likely Cause	Fix
'No session available' in DCV browser	DCV session not created or expired	Connect via EC2 Instance Connect and run: <code>dcv create-session mysession --owner ubuntu</code>
DCV browser shows blank page	Wrong IP address or port	Check current Public IP in EC2 console. Use port 8443.
SSM Ping Status: Offline	SSM agent stopped or IAM role issue	Run: <code>sudo snap restart amazon-ssm-agent</code> Wait 2 min and refresh page
SSM 'Could not assume role' error	IAM role trust policy or permissions wrong	Go to IAM → EC2-SSM-Role → fix trust policy, attach <code>AmazonSSMManagedInstanceCore</code>
systemctl: Unit not found	Ubuntu uses Snap for SSM agent	Use: <code>sudo snap restart amazon-ssm-agent</code> (not systemctl)
EC2 IP address changed	EC2 gives new IP on every start	Check new IP in EC2 console each time before connecting
Isaac Sim takes very long to load	First-time shader compilation or large scene	Normal on first load — wait 5–10 minutes. Subsequent loads are faster.
Pink or missing objects in scene	Broken USD file path reference	Re-open the .usd file or contact technical team for path repair
Low FPS / slow rendering	GPU memory pressure or heavy scene	Check GPU: <code>nvidia-smi</code> . Close other apps. Reduce render resolution if needed.

● 6. Key Concepts Explained Simply

What is Amazon EC2?

EC2 stands for Elastic Compute Cloud. Think of it as renting a powerful computer from Amazon that lives in their data center. You turn it on when you need it and turn it off when you don't — and you only pay for the time it is running. Our EC2 server has a high-performance GPU (graphics card) that can run the heavy 3D hospital simulation.

What is Amazon DCV?

DCV stands for Desktop Cloud Visualization. It is a technology that streams the screen, keyboard, and mouse of your cloud server to a web browser. Instead of sitting in front of the EC2 server physically, you open a browser and see the server's desktop as if it were right in front of you — with full graphics and real-time 3D rendering.

What is AWS SSM Session Manager?

SSM Session Manager is a secure way to open a command-line terminal on your EC2 server directly through the AWS Console in your browser — without needing SSH keys or opening firewall ports. Think of it as a secure backdoor into your server that only authorised AWS users can access.

What is an IAM Role?

IAM stands for Identity and Access Management. An IAM role is like an ID badge or security pass that you give to your EC2 server. When the server wears this badge, AWS knows who it is and what it is allowed to do. Without the badge, the server cannot access any AWS services — including SSM.

What is NVIDIA Isaac Sim / Omniverse?

NVIDIA Isaac Sim is a professional-grade 3D simulation platform built on top of NVIDIA Omniverse. It uses the same technology as Hollywood visual effects to create photorealistic environments with accurate physics, lighting, and material rendering. The USD (Universal Scene Description) file format — originally invented by Pixar — is used to store all the 3D scene data including models, materials, textures, and animations.

What is a Digital Twin?

A digital twin is a virtual replica of a real physical environment that is accurate enough to be used for real decision-making. Our hospital digital twin mirrors a real hospital space so closely that robots, workflows, and procedures can be tested in the virtual version before being deployed in the real one — eliminating risk and cost.

● 7. Current Project Status

Component	Status	Notes
Amazon EC2 Instance	■ Operational	Running Ubuntu, ap-south-1
Amazon DCV Remote Desktop	■ Active	Accessible via browser on port 8443
AWS SSM Session Manager	■ Connected	IAM role permanently fixed
SSM Agent (Snap)	■ Active & Enabled	Correct Ubuntu Snap installation
IAM Role (EC2-SSM-Role)	■ Correctly Configured	Trust policy + permissions both set
Hospital OT Environment	■ Built & Live	Full Operating Theatre scene complete
Patient Room Environment	■ Built & Live	Full patient room scene complete
Spot Robot in Isaac Sim	■ In Progress	Currently configuring sensors
Sensor Configuration (Spot)	■ In Progress	Active work item

Next Steps

- Complete sensor configuration on the Spot robot in Isaac Sim
- Integrate sensor data output with ROS2 or Python API
- Test robot navigation within the hospital simulation environment
- Implement automated DCV session creation on server startup
- Set up CloudWatch monitoring and alerts for the EC2 instance

End of Document

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